

Prompt Engineering, AI Agent & AI App Development with Claude

Course Overview

Pelatihan ini dirancang untuk menggabungkan kemampuan **Prompt Engineering**, **AI Agent Development**, dan **AI Application Development** menggunakan Claude sebagai Large Language Model (LLM). Program ini berfokus pada bagaimana membangun solusi AI modern mulai dari desain prompt, AI automation, AI Agent, hingga integrasi aplikasi berbasis AI untuk kebutuhan bisnis dan enterprise.

Peserta akan mempelajari:

1. Fundamental Large Language Model (LLM)
2. Teknik Prompt Engineering profesional
3. AI Agent workflow & orchestration
4. Claude API integration
5. AI application development
6. Retrieval Augmented Generation (RAG)
7. Multi-step AI automation
8. Integrasi AI dengan tools, database, dan workflow bisnis
9. Capstone Project implementasi AI end-to-end

Program ini menitikberatkan pada pendekatan hands-on dan studi kasus nyata sehingga peserta dapat langsung mengimplementasikan AI solution di lingkungan kerja.

Learning Objectives

Setelah mengikuti pelatihan ini, peserta mampu:

1. Memahami konsep dan arsitektur Large Language Model (LLM)
2. Mendesain prompt profesional dan efektif
3. Mengoptimalkan output AI menggunakan advanced prompting techniques
4. Menggunakan Claude API untuk pengembangan aplikasi AI
5. Membangun AI application berbasis Claude
6. Mengembangkan AI Agent untuk automation workflow
7. Mengimplementasikan RAG (Retrieval Augmented Generation)
8. Mengintegrasikan AI dengan tools, database, dan API eksternal

9. Mendesain AI use case untuk kebutuhan bisnis
10. Mengembangkan solusi AI end-to-end secara scalable

Duration

5 Days (40 Hours)

Target Audience

Program ini ditujukan untuk:

1. Software Developers
2. Backend / Frontend Engineers
3. AI / ML Engineers (Beginner–Intermediate)
4. Data Analysts / Data Engineers
5. Product Managers / Product Owners
6. Digital Transformation Team
7. Innovation Team
8. IT Solution Architect
9. Startup / Internal Innovation Lab
10. Business Technology Team

Prerequisites

Peserta disarankan memiliki:

1. Pemahaman dasar AI / ChatGPT / Claude
2. Basic programming (Python / JavaScript)
3. Pemahaman dasar API & aplikasi web
4. Familiar dengan workflow digital bisnis

Detailed Syllabus (5 Days – 40 Hours)

DAY 1 : Prompt Engineering Fundamentals

Module 1: Introduction to Large Language Model (Claude)

- Fundamental konsep LLM
- Cara kerja Generative AI
- Transformer architecture overview
- Tokenization & context window
- Claude capabilities & limitations
- AI reasoning & hallucination

Module 2: Prompt Engineering Basics

- Anatomy of effective prompt
- Role prompting
- Context engineering
- Output formatting
- Instruction design
- Prompt template structure

Module 3: Prompting Techniques

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought prompting
- Step-by-step reasoning
- Persona-based prompting
- Structured prompting

Hands-on Lab

- Create effective prompts
- Prompt optimization practice
- Output quality testing

DAY 2 : Advanced Prompt Engineering & AI Workflow

Module 4: Structured Output & Prompt Optimization

- JSON output generation
- Controlled responses
- Prompt refinement
- Prompt testing strategy
- Error handling
- Prompt evaluation framework

Module 5: Prompt for Business Use Cases AI for customer service

- AI for document automation
- AI for report generation
- AI for data analysis
- AI for internal knowledge assistant

Module 6: AI Workflow Automation

- Multi-step AI workflow
- AI orchestration
- Tool integration concept
- AI automation pipeline
- AI task chaining

Hands-on Lab

- Build production-ready prompts
- Create AI workflow automation

DAY 3 : AI Agent Development with Claude

Module 7: Introduction to AI Agent

- AI Agent fundamentals
- AI Agent vs Chatbot
- Goal-oriented AI systems
- Agent workflow architecture
- Planning & reasoning process
- AI memory concept

Module 8: AI Agent Orchestration

- Single-agent architecture
- Multi-agent system
- Tool calling / function calling
- Agent execution flow
- Decision making & action planning
- AI task delegation

Module 9: Building AI Agent with Claude API

- Claude API integration
- Authentication & API management
- Backend integration
- Input/output handling
- Agent interaction flow
- AI Agent deployment basics

Hands-on Lab

- Build AI Agent assistant
- Create automation agent
- Integrate AI Agent with external tools

DAY 4 : AI App Development & RAG

Module 10: Build AI Application

- AI application architecture
- Frontend integration
- Chat interface development
- Backend AI integration
- Session & context management

Module 11: Retrieval Augmented Generation (RAG)

- RAG architecture
- Embeddings fundamentals
- Vector database concept Semantic search
- Knowledge retrieval workflow
- Document ingestion pipeline

Module 12: Advanced AI Applications

- AI internal assistant
- AI knowledge management
- AI automation system
- Performance optimization
- AI cost optimization
- Scaling AI applications

Hands-on Lab

- Build RAG-based AI app
- Build enterprise AI assistant

DAY 5 : AI Product, Governance & Capstone Project

Module 13: AI Product Design

- AI use case selection
- Business value identification
- AI solution architecture
- AI product roadmap
- User experience for AI systems

Module 14: AI Governance, Risk & Security

- AI ethics & governance
- AI bias & fairness
- AI security risk
- Prompt injection awareness
- Data privacy & compliance
- Responsible AI implementation

Capstone Project

Participants will:

1. Design AI use case
2. Build AI application
3. Build AI Agent
4. Integrate Claude API
5. Apply prompt engineering
6. Implement RAG
7. Present final solution

Tools & Technologies

1. Claude (LLM)
2. Claude API
3. Python / JavaScript
4. Vector Database
5. LangChain / Agent Framework (optional)
6. REST API Integration
7. AI Workflow Tools

Training Methodology

1. 30% Concept
2. 60% Hands-on Lab
3. 5% Case Study
4. 5% Discussion

Metode pelatihan difokuskan pada implementasi langsung menggunakan studi kasus nyata dan project-based learning.

Deliverables

Peserta akan mendapatkan:

1. Training Materials
2. Prompt Engineering Templates
3. AI Agent Templates
4. AI App Starter Code
5. API Integration Guide
6. Use Case Examples
7. Hands-on Lab Files
8. Capstone Project Files
9. Certificate of Completion

Expected Outcome

Setelah mengikuti pelatihan, peserta dapat:

1. Mendesain prompt profesional
2. Membangun AI application
3. Mengembangkan AI Agent
4. Mengintegrasikan AI ke sistem bisnis
5. Mengembangkan AI automation workflow
6. Mengimplementasikan RAG
7. Mengoptimalkan performa AI
8. Mendesain solusi AI enterprise
9. Mengembangkan use case AI end-to-end

Key Value for Organization

1. Mempercepat implementasi AI
2. Mendukung digital transformation
3. Mengurangi biaya automation development
4. Meningkatkan productivity & innovation
5. Memanfaatkan AI secara maksimal
6. Meningkatkan efisiensi operasional

Relevant Business Use Cases

1. AI Customer Service Agent
2. AI Internal Assistant
3. AI Knowledge Management System
4. AI Document Automation
5. AI Data Analysis Assistant
6. AI Workflow Automation
7. AI Helpdesk Assistant
8. AI Reporting Automation
9. Enterprise AI Chatbot
10. AI Decision Support System